INCH-POUND

A–A–59590/3B
7 April 2006
SUPERSEDING
A-A-59590/3A
21 June 2002

COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE, .225 X .260 INCH BODY SIZE, WITH VISUAL LOCK INDICATION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART OR IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see NOTES).

AA59590	/03	*	*	* *	W, D, E
CID number	Applicable CID specification sheet	Finish	Mounting	Length (see table I)	Options

Example: AA59590/03PK63 is the PIN for a nickel finished, 6.3 inch (160.0 mm) long card holder with a retained actuating screw. The card holder also has three counterbored through holes for use with rivet fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position. When the card holder is installed properly, it is capable of withstanding 60g/6ms of shock, 25 G–rms of vibration, and provides from 2 to 4 degrees C/W/inch thermal resistance transfer between the circuit card assembly and the heat sink surfaces.

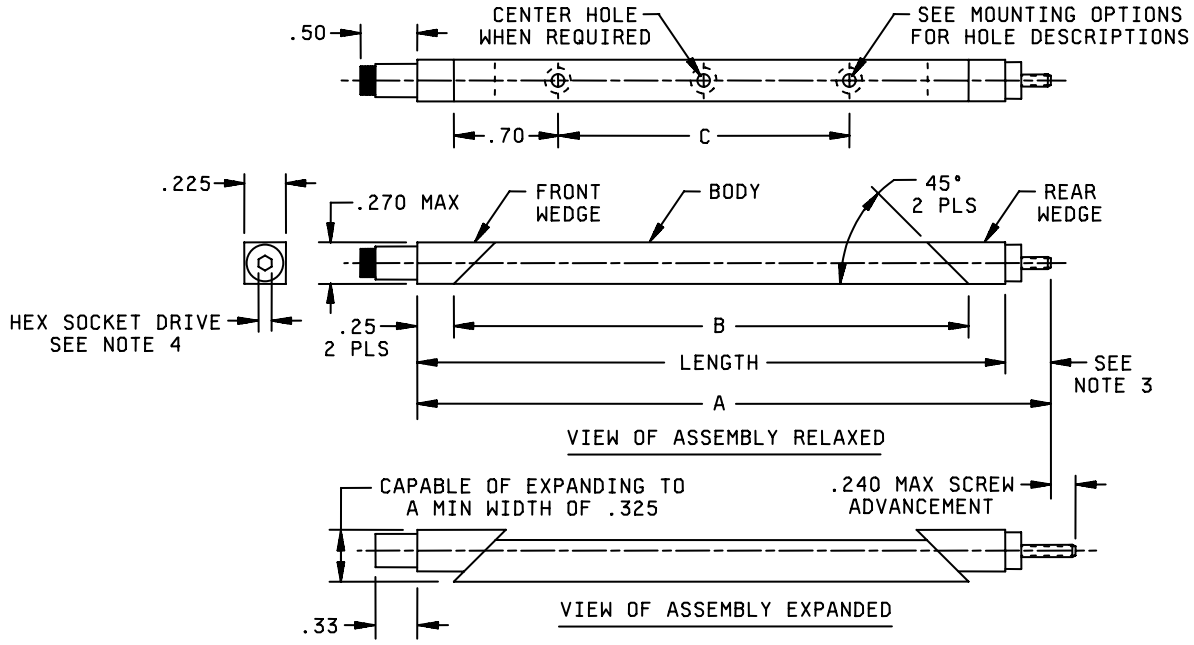
Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified on figures 1, 2, 3, 4, in table I, and A–A–59590.

Visual lock indicator (see figure 2). Card holders shall have a visual indicator to show when the card holder is in its relaxed (unlocked) state. When the card holder is in the unlocked (relaxed) state, the actuating end of the screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened (assembly expanded) so that the assembly is in the locked position, this red band shall be concealed.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 6 inch-pounds (0.68 N-m) for assemblies using no options or option “W” and 7.2 inch-pounds (0.81 N-m) for assemblies using options “E” or “D”.

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .300 inch (7.62 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.



Inches	mm	Inches	mm	Inches	mm	Inches	mm	Inches	mm
.20	5.1	.240	6.1	.270	6.9	.33	8.4	.70	17.8
.225	5.7	.25	6.4	.325	8.3	.50	12.7		

NOTES:

1. Dimensions are in inches. Millimeters are given for information only.
2. Unless otherwise specified, tolerances are for ± .02 inch (0.5 mm) for two place decimals and ± .010 (0.25 mm) for three place decimals.
3. Dimension shall be .20 (5.1 mm) maximum on card holders without option “W” or “D” and .30 (7.6 mm) maximum on card holders with option “W” or “D”.
4. The across flats dimension for hex drive socket shall be .094 inch (2.38 mm) for mounting options “J”, “K”, “N”, “T”, “R”, “V”, and “S”. The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (.098 inch) across flats for mounting options “G” and “M”.

FIGURE 1. Relaxed and expanded dimensions.

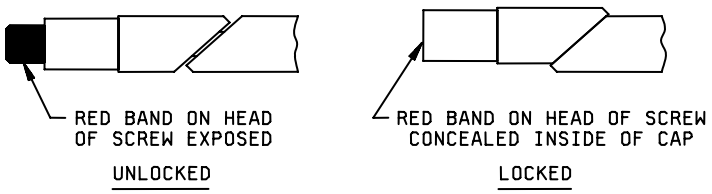


FIGURE 2 Visual lock indicator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE I. Additional assembly dimensions (see figure 1). 1/

PIN length designator	Dimension "LENGTH" ± .02 (0.5)	Dimension "A" +.02 / –.09 (+0.5 / –2.3)		Dimension "B" ± .03 (0.8)	Dimension "C" ± .005 (0.13)
		Without options W or D	With options W or D		
18	1.8 (45.7)	2.20 (55.9)	2.30 (58.4)	1.30 (33.0)	.75 (19.1)
23	2.3 (58.4)	2.70 (68.6)	2.80 (71.1)	1.80 (45.7)	.40 (10.2)
28	2.8 (71.1)	3.20 (81.3)	3.30 (83.8)	2.30 (58.4)	.90 (22.9)
30	3.0 (76.2)	3.40 (86.4)	3.50 (88.9)	2.50 (63.5)	1.10 (27.9)
33	3.3 (83.8)	3.70 (94.0)	3.80 (96.5)	2.80 (71.1)	1.40 (35.6)
35	3.5 (88.9)	3.90 (99.1)	4.-0 (101.6)	3.00 (76.2)	1.60 (40.6)
38	3.8 (96.5)	4.20 (106.7)	4.30 (109.2)	3.30 (83.8)	1.90 (48.3)
43	4.3 (109.2)	4.70 (119.4)	4.80 (121.9)	3.80 (96.5)	2.40 (61.0)
48	4.8 (121.9)	5.20 (132.1)	5.30 (134.6)	4.30 (109.2)	2.90 (73.7)
53	5.3 (134.6)	5.70 (144.8)	5.80 (147.3)	4.80 (121.9)	3.40 (86.4)
55	5.5 (139.7)	5.90 (149.9)	6.00 (152.4)	5.00 (127.0)	3.60 (91.4)
58	5.8 (147.3)	6.20 (157.5)	6.30 (160.0)	5.30 (134.6)	3.90 (99.1)
61	6.1 (154.9)	6.50 (165.1)	6.60 (167.6)	5.60 (142.2)	4.20 (106.7)
63	6.3 (160.0)	6.70 (170.2)	6.80 (172.7)	5.80 (147.3)	4.40 (111.8)
67	6.7 (170.2)	7.10 (180.3)	7.20 (182.9)	6.20 (157.5)	4.80 (121.9)

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for information only.

Finish. The finish options shall be as specified in A–A–59590. The finish options available for this specification sheet are as follows: "B" (black anodize), "C" (chemical film), "E" (electroless nickel), "H" (hard black anodize), and "P" (electrodeposited nickel).

Mounting options. The mounting options shall be as specified in A–A–59590. The mounting options available for this specification sheet are as follows: "J" or "K" (rivet mounting holes), "N" (no mounting holes), "R" or "T" (tapped 0-80 UNF 2B holes), "V" or "S" (tapped 2-56 UNC 2B holes), or "G" or "M" (tapped M2.5 x.45 metric holes). Card holders using mounting option "J" or "K" may be shipped unassembled.

Mounting holes (when required). Two mounting holes are required on card holders less than 5.30 inches (134.6 mm) in length. Three mounting holes are required on card holders 5.30 inches (134.6 mm) or greater in length. The third mounting hole shall be centered on the mounting body. See figure 1 and table I for mounting hole locations and spacing requirements.

Rivet mounting holes. The holes used for rivet mounting shall be .066 to .073 inch (1.68 to 1.85 mm) diameter through holes, countersunk 100 degrees by .140 inch (3.56 mm) diameter with an access/clearance counterbore hole of .156 inch (3.96 mm) diameter by .200 inch (5.08 mm) deep.

Length, expanded, and relaxed dimensions. The length, expanded, and relaxed dimensions shall be as specified on figure 1. The length designator shall be as specified in A–A–59590 and the available lengths for this specification sheet are listed in table I.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

Options. Card holders can have options “W” (lockwasher and flat washer), “E” (screw self-locking element), or “D” (combination of options “W” and “E”) as described below. Card holders not requiring these options shall leave the options position in the PIN blank.

Lockwasher and flat washer (see figure 3). A lockwasher and flat washer located under the screw head will provide for additional resistance to loosening from shock and vibration. Card holders requiring a lockwasher and flat washer option shall include a suffix “W” in the PIN (see classification and notes).

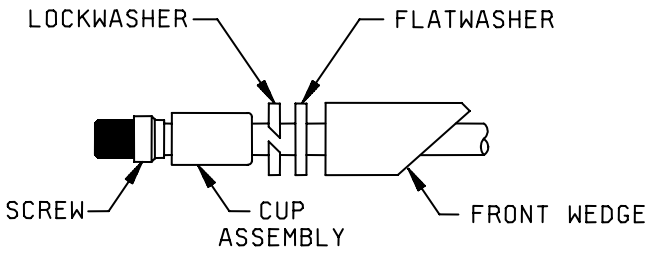


FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element will provide prevailing torque for resistance to loosening from shock and vibration. Card holders requiring a screw self-locking element shall include a suffix “E” in the PIN (see classification and notes).

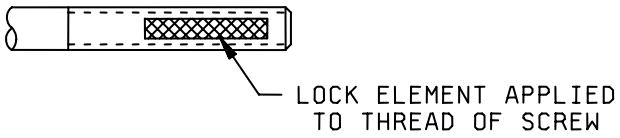


FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix “D” in the PIN (see classification and notes).

NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at http://assist.daps.dla.mil/quicksearch/ or http://assist.daps.dla.mil/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

Manufacturer CAGE	Manufacturer name and address	Manufacturer contact information
18915	Birtcher Pentair Electronic Packaging 14100 Danielson Street Poway, CA 92064–6898	Telephone: (858) 679–4550 Facsimile: (858) 679–4555 Electronic mail: infobirtcher@pentair-ep.com URL: www.birtcherproducts.com
52094	Calmark Corporation 4915 Walnut Grove Avenue San Gabriel, CA 91776–2099	Telephone: (626) 287–0451 Facsimile: (626) 287–7350 Electronic mail: sales@calmark.com URL: www.calmark.com

Part number supersession data. These CID specification sheet part numbers supersede the following manufacturer's part numbers in table II and on DSCC Drawing 84103 (table III) as shown. The CID PINs listed in table II are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.

TABLE II. Commercial part number supersession data.

PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 52094
CJ28	40VI–6	CJ28	V225CR2.80H
CN28	40VI–6–A	CN28	V225CR2.80
CT28	40VI–6–T	CT28	V225CR2.80T0
CV28	40VI–6–S	CV28	V225CR2.80T2
CG28	40VI–6–M	CG28	MV225CR2.80TM2.5
CJ28E	40VI–6–L	CJ28E	V225CR2.80HL
CN28E	40VI–6–A–L	CN28E	V225CR2.80L
CT28E	40VI–6–T–L	CT28E	V225CR2.80T0L
CV28E	40VI–6–S–L	CV28E	V225CR2.80T2L
CG28E	40VI–6–M–L	CG28E	MV225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 52094
CJ28W	40VI–6–LF	CJ28W	WV225CR2.80H
CN28W	40VI–6–LF–A	CN28W	WV225CR2.80
CT28W	40VI–6–LF–T	CT28W	WV225CR2.80T0
CV28W	40VI–6–LF–S	CV28W	WV225CR2.80T2
CG28W	40VI–6–LF–M	CG28W	MWV225CR2.80TM2.5
CJ28D	40VI–6–LF–L	CJ28D	WV225CR2.80H
CN28D	40VI–6–LF–A–L	CN28D	WV225CR2.80
CT28D	40VI–6–LF–T–L	CT28D	WV225CR2.80T0
CV28D	40VI–6–LF–S–L	CV28D	WV225CR2.80T2
CG28D	40VI–6–LF–M–L	CG28D	MWV225CR2.80TM2.5
BJ28	40VI–6–B	BJ28	VA225CR2.80H
BN28	40VI–6–B–A	BN28	VA225CR2.80
BT28	40VI–6–B–T	BT28	VA225CR2.80T0
BV28	40VI–6–B–S	BV28	VA225CR2.80T2
BG28	40VI–6–B–M	BG28	MVA225CR2.80TM2.5
BJ28E	40VI–6–B–L	BJ28E	VA225CR2.80HL
BN28E	40VI–6–B–A–L	BN28E	VA225CR2.80L
BT28E	40VI–6–B–T–L	BT28E	VA225CR2.80T0L
BV28E	40VI–6–B–S–L	BV28E	VA225CR2.80T2L
BG28E	40VI–6–B–M–L	BG28E	MVA225CR2.80TM2.5L
BJ28W	40VI–6–B–LF	BJ28W	WVA225CR2.80H
BN28W	40VI–6–B–LF–A	BN28W	WVA225CR2.80
BT28W	40VI–6–B–LF–T	BT28W	WVA225CR2.80T0
BV28W	40VI–6–B–LF–S	BV28W	WVA225CR2.80T2
BG28W	40VI–6–B–LF–M	BG28W	MWVA225CR2.80TM2.5
BJ28D	40VI–6–B–LF–L	BJ28D	WVA225CR2.80HL
BN28D	40VI–6–B–LF–A–L	BN28D	WVA225CR2.80L
BT28D	40VI–6–B–LF–T–L	BT28D	WVA225CR2.80T0L
BV28D	40VI–6–B–LF–S–L	BV28D	WVA225CR2.80T2L
BG28D	40VI–6–B–LF–M–L	BG28D	MWVA225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 52094
HJ28	40VI–6–B3	HJ28	VHA225CR2.80H
HN28	40VI–6–B3–A	HN28	VHA225CR2.80
HT28	40VI–6–B3–T	HT28	VHA225CR2.80T0
HV28	40VI–6–B3–S	HV28	VHA225CR2.80T2
HG28	40VI–6–B3–M	HG28	MVHA225CR2.80TM2.5
HJ28E	40VI–6–B3–L	HJ28E	VHA225CR2.80HL
HN28E	40VI–6–B3–A–L	HN28E	VHA225CR2.80L
HT28E	40VI–6–B3–T–L	HT28E	VHA225CR2.80T0L
HV28E	40VI–6–B3–S–L	HV28E	VHA225CR2.80T2L
HG28E	40VI–6–B3–M–L	HG28E	MVHA225CR2.80TM2.5L
HJ28W	40VI–6–B3–LF	HJ28W	WVHA225CR2.80H
HN28W	40VI–6–B3–LF–A	HN28W	WVHA225CR2.80
HT28W	40VI–6–B3–LF–T	HT28W	WVHA225CR2.80T0
HV28W	40VI–6–B3–LF–S	HV28W	WVHA225CR2.80T2
HG28W	40VI–6–B3–LF–M	HG28W	MWVHA225CR2.80TM2.5
HJ28D	40VI–6–B3–LF–L	HJ28D	WVHA225CR2.80HL
HN28D	40VI–6–B3–LF–AL	HN28D	WVHA225CR2.80L
HT28D	40VI–6–B3–LF–TL	HT28D	WVHA225CR2.80T0L
HV28D	40VI–6–B3–LF–SL	HV28D	WVHA225CR2.80T2L
HG28D	40VI–6–B3–LF–ML	HG28D	MWVHA225CR2.80TM2.5L
PJ28	40VI–6–N	EJ28	VEN225CR2.80H
PN28	40VI–6–N–A	EN28	VEN225CR2.80
PT28	40VI–6–N–T	ET28	VEN225CR2.80T0
PV28	40VI–6–N–S	EV28	VEN225CR2.80T2
PG28	40VI–6–N–M	EG28	MVEN225CR2.80TM2.5
PJ28E	40VI–6–N–L	EJ28E	VEN225CR2.80HL
PN28E	40VI–6–N–A–L	EN28E	VEN225CR2.80L
PT28E	40VI–6–N–T–L	ET28E	VEN225CR2.80T0L
PV28E	40VI–6–N–S–L	EV28E	VHEN25CR2.80T2L
PG28E	40VI–6–N–M–L	EG28E	MVEN225CR2.80TM2.5L

See footnote at end of table.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE II. Commercial part number supersession data – Continued.

PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 18915	PIN designator AA59590/03	Vendor similar designator or type part number 1/ CAGE 52094
PJ28W	40VI–6–N–LF	EJ28W	WVEN225CR2.80H
PN28W	40VI–6–N–LF–A	EN28W	WVEN225CR2.80
PT28W	40VI–6–N–LF–T	ET28W	WVEN225CR2.80T0
PV28W	40VI–6–N–LF–S	EV28W	WVEN225CR2.80T2
PG28W	40VI–6–N–LF–M	EG28W	MWVEN225CR2.80TM2.5
PJ28D	40VI–6–N–LF–L	EJ28D	WVEN225CR2.80HL
PN28D	40VI–6–N–LF–A–L	EN28D	WVEN225CR2.80L
PT28D	40VI–6–N–LF–T–L	ET28D	WVEN225CR2.80T0L
PV28D	40VI–6–N–LF–S–L	EV28D	WVEN225CR2.80T2L
PG28D	40VI–6–N–LF–M–L	EG28D	MWVEN225CR2.80TM2.5L

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE III. Supersession information.

Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN	Superseded (old) PIN in accordance with DSCC Drawing 84103	Superseding (new) CID PIN
84103–08B	AA59590/03BJ28	84103–08C	AA59590/03CJ28
84103–09B	AA59590/03BJ30	84103–09C	AA59590/03CJ30
84103–10B	AA59590/03BJ35	84103–10C	AA59590/03CJ35
84103–11B	AA59590/03BJ38	84103–11C	AA59590/03CJ38
84103–12B	AA59590/03BJ48	84103–12C	AA59590/03CJ48
84103–13B	AA59590/03BJ53	84103–13C	AA59590/03CJ53
84103–14B	AA59590/03BJ58	84103–14C	AA59590/03CJ58

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.

TABLE IV. Example of PIN with available length designators.

PIN designator AA59590/03	Vendor similar designator or type part number 1/ 2/ CAGE 18915	PIN designator AA59590/03	Vendor similar designator or type part number 1/ 2/ CAGE 52094
PJ18	40VI–4–N	EJ18	VEN225CR1.80H
PJ23	40VI–5–N	EJ23	VEN225CR2.30H
PJ28	40VI–6–N	EJ28	VEN225CR2.80H
PJ30	3/	EJ30	VEN225CR3.00H
PJ33	40VI–7–N	EJ33	VEN225CR3.30H
PJ35	3/	EJ35	VEN225CR3.50H
PJ38	40VI–8–N	EJ38	VEN225CR3.80H
PJ43	40VI–9–N	EJ43	VEN225CR4.30H
PJ48	40VI–10–N	EJ48	VEN225CR4.80H
PK53	40VI–11–N	EK53	VEN225CR5.30EH
PK55	3/	EK55	VEN225CR5.50EH
PK58	40VI–12–N	EK58	VEN225CR5.80EH
PK61	3/	EK61	VEN225CR6.10EH
PK63	40VI–13–N	EK63	VEN225CR6.30EH
PK67	3/	EK67	VEN225CR6.70EH

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.
3/ Consult manufacturer for part number.

Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 11
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FSS

Preparing Activity:
DLA – CC

Project 5998–2005–012

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at http://assist.daps.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T10:19Z
Check the source to verify that this is the current version before use.